



Inspiring Excellence

Department of Mathematics and Natural Sciences

Mid-Term Examination

Semester: Spring 2025

Course Title: MATHEMATICS I: Differential Calculus and Coordinate Geometry

Course Code: MAT- 110

Time: 1 hour 10 minutes

Date: March 19, 2025

Instructions:

Total marks: 40

[5 × 8 = 40]

- Answer any **five** questions.
 - The figures in the right margin indicate full marks.
 - The total marks will be converted to 20.
 - All your works must be inside the answer script. Otherwise, it will not be counted.
-

1. Evaluate limit: [8]

$$\lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}}$$

2. Test the differentiability and continuity of the following function at $x = 0$. [8]

$$f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x \leq 0 \\ 3 - 2x, & 0 < x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$$

3. Find $\frac{dy}{dx}$ where $y = (\sin(e^x))^{\ln(e^{2x})}$ [8]

4. Find the open intervals on which the function [8]
 $f(x) = x^4 - 4x^3 + 4x^2 + 3$ is increasing and decreasing. Also find the relative extrema.

5. Verify the statement of the Mean Value Theorem for the function [8]
 $f(x) = x^3 - x^2 + 9x - 2$
over the interval $[1, 4]$, and find all values of 'c' in that interval that satisfy the conclusion of the theorem.

6. Find the Maclaurin polynomials P_0, P_1, P_2, P_3 of the function $f(x) = \ln\left(\frac{1+x}{1-x}\right)$. [8]

Hint: Use the formula $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$.

SOLUTION & MARKING SCHEME
(Created by: SAKAL ROY [SKY])

TO APPRECIATE YOUR MATHEMATICAL THINKING, ANY VALID MATHEMATICAL WAY OF SOLVING WILL BE TREATED EQUALLY AND ASSESSED ACCORDINGLY, AS LONG AS IT'S CONSISTENT WITH THE PROVIDED INSTRUCTIONS OF THE QUESTION.

IN THIS ASSESSMENT PROCESS EVERY PROBLEM WILL BE ASSESSED BASED ON 3 CRITERIA:

- A. MATHEMATICAL THINKING
 - B. MATHEMATICAL KNOWLEDGE
 - C. MATHEMATICAL SKILL
-

1. Evaluate limit:

[8]

$$\lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}}$$

Solution

<https://www.emathhelp.net/calculators/calculus-1/limit-calculator/?f=%281%2Bsin+%28x%29%29%5E%281%2Fx%29&var=&val=0&dir=>

Since $u = e^{\ln(u)}$, then:

$$\lim_{x \rightarrow 0} (\sin(x) + 1)^{\frac{1}{x}} = \lim_{x \rightarrow 0} e^{\ln((\sin(x)+1)^{\frac{1}{x}})}$$

Simplify:

$$\lim_{x \rightarrow 0} e^{\ln((\sin(x)+1)^{\frac{1}{x}})} = \lim_{x \rightarrow 0} e^{\frac{\ln(\sin(x)+1)}{x}}$$

Move the limit under the exponential:

$$\lim_{x \rightarrow 0} e^{\frac{\ln(\sin(x)+1)}{x}} = e^{\lim_{x \rightarrow 0} \frac{\ln(\sin(x)+1)}{x}}$$

Since we have an indeterminate form of type $\frac{0}{0}$, we can apply the l'Hopital's rule:

$$e^{\lim_{x \rightarrow 0} \frac{\ln(\sin(x)+1)}{x}} = e^{\lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\ln(\sin(x)+1))}{\frac{d}{dx}(x)}}$$

For steps, see [derivative calculator](#).

$$e^{\lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\ln(\sin(x)+1))}{\frac{d}{dx}(x)}} = e^{\lim_{x \rightarrow 0} \frac{\cos(x)}{\sin(x)+1}}$$

Substitute the variable with the value:

$$e^{\lim_{x \rightarrow 0} \frac{\cos(x)}{\sin(x)+1}} = e^1$$

Therefore,

$$\lim_{x \rightarrow 0} (\sin(x) + 1)^{\frac{1}{x}} = e$$

Answer: $\lim_{x \rightarrow 0} (\sin(x) + 1)^{\frac{1}{x}} = e$

2. Test the differentiability and continuity of the following function at $x = 0$.

[8]

$$f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x \leq 0 \\ 3 - 2x, & 0 < x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$$

Solution
Differentiability
To test the differentiability of $f(x)$ at $x = a$, We've to explore the following:
$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{f(a) - f(x)}{a - x} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{f(a) - f(a - h)}{h}$
Here, let, $f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x \leq 0 \\ 3 - 2x, & 0 < x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$ and $a = 0$
So, We've to verify:

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(0) - f(x)}{0 - x} = \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(0) - f(0 - h)}{h}$$

$$\text{Since, } f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x \leq 0 \\ 3 - 2x, & 0 < x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$$

$$\text{So, } f(0) = 3 + 2 \cdot 0 = 3$$

$$\Rightarrow f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - 3}{x - 0} = \lim_{h \rightarrow 0} \frac{f(0 + h) - 3}{h} = \lim_{h \rightarrow 0} \frac{3 - f(0 - h)}{h}$$

First we've to find: Right Hand Derivative (R. H. D)

$$f'_+(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - 3}{x - 0} = \lim_{h \rightarrow 0^+} \frac{f(0 + h) - 3}{h} = \lim_{h \rightarrow 0^+} \frac{3 - f(0 - h)}{h}$$

$$\lim_{x \rightarrow 0^+} \frac{f(x) - 3}{x - 0}$$

$$\text{Now, since, } f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x \leq 0 \\ 3 - 2x, & 0 < x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases} \text{ and } 0^+ > 0$$

$$\text{So, for } x \rightarrow 0^+, f(x) = 3 - 2x$$

$$\text{So, } \lim_{x \rightarrow 0^+} \frac{f(x) - 3}{x - 0}$$

$$= \lim_{x \rightarrow 0^+} \frac{3 - 2x - 3}{x - 0}$$

$$= \lim_{x \rightarrow 0^+} \frac{-2x}{x}$$

$$= \lim_{x \rightarrow 0^+} (-2)$$

$$= -2$$

$$\text{So, } f'_+(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - 3}{x - 0} = -2$$

Now we've to find: Left Hand Derivative (L. H. D)

$$f'_-(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - 3}{x - 0} = \lim_{h \rightarrow 0^-} \frac{f(0 + h) - 3}{h} = \lim_{h \rightarrow 0^-} \frac{3 - f(0 - h)}{h}$$

$$\lim_{x \rightarrow 0^-} \frac{f(x) - 3}{x - 0}$$

$$\text{Now, since, } f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x \leq 0 \\ 3 - 2x, & 0 < x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases} \text{ and } 0^- < 0$$

$$\text{So, for } x \rightarrow 0^-, f(x) = 3 + 2x$$

$$\text{So, } \lim_{x \rightarrow 0^-} \frac{f(x) - 3}{x - 0}$$

$$= \lim_{x \rightarrow 0^-} \frac{3 + 2x - 3}{x - 0}$$

$$= \lim_{x \rightarrow 0^-} \frac{2x}{x}$$

$$= \lim_{x \rightarrow 0^-} (2)$$

$$= 2$$

$$\text{So, } f'_-(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - 3}{x - 0} = 2$$

$$\text{Since, } \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} \neq \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0}, \text{ so, } f'_+(0) \neq f'_-(0)$$

$$\text{Therefore, } f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \text{ doesn't exist.}$$

$$\text{So, } f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x \leq 0 \\ 3 - 2x, & 0 < x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases} \text{ is not differentiable at } x = 0.$$

Continuity

To test the **continuity** of **f(x)** at **x = a**, We've to explore the following:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Here, **a = 0**.

Check the continuity of f(x) at x = 0 by verifying $\lim_{x \rightarrow 0} f(x) = f(0)$.

You can do that by finding the values of $\lim_{x \rightarrow 0^+} f(x)$, $\lim_{x \rightarrow 0^-} f(x)$ and f(0) separately.

We've to find: $\lim_{x \rightarrow 0} f(x)$

First we've to find: Right Hand Limit (R. H. L) = $\lim_{x \rightarrow 0^+} f(x)$

Method 1:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (3 - 2x)$$

Since, g(x) = (3 - 2x),
is a polynomial and continuous at any value of x.

$$\text{So, } \lim_{x \rightarrow 0^+} g(x) = g(0)$$

$$\text{or, } \lim_{x \rightarrow 0^+} (3 - 2x) = 3 - 2 \cdot 0 = 3$$

Method 2:

$$\lim_{x \rightarrow 0^+} f(x)$$

Let, x = 0 + h, where h → 0⁺

$$\text{So, } \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{h \rightarrow 0^+} f(0 + h)$$

Now, 0 + h > 0 since h → 0⁺

$$\text{So, } f(x) = 3 - 2x$$

$$\text{So, } \lim_{h \rightarrow 0^+} f(h)$$

$$= \lim_{h \rightarrow 0^+} (3 - 2h)$$

$$\text{Since, } g(h) = (3 - 2h),$$

is a polynomial and continuous at any value of h.

$$\text{So, } \lim_{h \rightarrow 0^+} g(h) = g(0)$$

$$\text{or, } \lim_{x \rightarrow 0^+} (3 - 2x) = g(0) = 3 - 2 \cdot 0 = 3$$

Now, we've to find: Left Hand Limit (L. H. L) = $\lim_{x \rightarrow 0^-} f(x)$

Method 1:

$$\lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{x \rightarrow 0^-} (3 + 2x)$$

Since, g(x) = (3 + 2x),
is a polynomial and continuous at any

Method 2:

$$\lim_{x \rightarrow 0^-} f(x)$$

Let, x = 0 - h, where h → 0⁺

$$\text{So, } \lim_{x \rightarrow 0^-} f(x)$$

value of x . So, $\lim_{x \rightarrow 0^-} g(x) = g(0)$ or, $\lim_{x \rightarrow 0^-} (3 + 2x) = 3 + 2 \cdot 0 = 3$	$= \lim_{h \rightarrow 0^+} f(0 - h)$ Now, $0 - h > 0$ since $h \rightarrow 0^+$ So, $f(x) = (3 + 2x)$ So, $\lim_{h \rightarrow 0^+} f(-h)$ $= \lim_{h \rightarrow 0^+} (3 + 2(-h))$ Since, $g(h) = (3 - 2h)$, is a polynomial and continuous at any value of h . So, $\lim_{h \rightarrow 0^+} g(h) = g(0)$ or, $\lim_{x \rightarrow 0^+} (3 + 2x) = g(0) = 3 + 2 \cdot 0 = 3$
Now, $f(0) = 3$.	
If $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a)$ then $f(x)$ is continuous at $x = a$. And if all of $\lim_{x \rightarrow a^+} f(x)$, $\lim_{x \rightarrow a^-} f(x)$ and $f(a)$ aren't equal then $f(x)$ is not continuous at $x = a$ Here, $a = 0$, then: $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = f(0)$. So, $f(x)$ is continuous at $x = 0$.	

3. Find $\frac{dy}{dx}$ where $y = (\sin(e^x))^{\ln(e^{2x})}$

[8]

Solution
See full solution here: https://www.emathhelp.net/calculators/calculus-1/derivative-calculator/?f=%28sin+%28e%5Ex%29%29+%5E+%28ln+%28e%5E%282x%29%29%29&order=1&var=x&p=
Initially a bit of algebraic manipulation helps converting the problem into an easier one. $y = (\sin(e^x))^{\ln(e^{2x})} \Rightarrow y = (\sin(e^x))^{2x \ln(e)} = (\sin(e^x))^{2x \cdot 1} = (\sin(e^x))^{2x}$
Use the formula $f^{g(x)}(x) = e^{g(x) \ln(f(x))}$ with $f(x) = \sin(e^x)$ and $g(x) = 2x$ to rewrite the complex expression: $\left(\frac{d}{dx} (\sin^{2x}(e^x)) \right) = \left(\frac{d}{dx} (e^{2x \ln(\sin(e^x))}) \right)$ The function $e^{2x \ln(\sin(e^x))}$ is the composition $f(g(x))$ of two functions $f(u) = e^u$ and $g(x) = 2x \ln(\sin(e^x))$. Apply the chain rule $\frac{d}{dx} (f(g(x))) = \frac{d}{du} (f(u)) \frac{d}{dx} (g(x))$: $\left(\frac{d}{dx} (e^{2x \ln(\sin(e^x))}) \right) = \left(\frac{d}{du} (e^u) \frac{d}{dx} (2x \ln(\sin(e^x))) \right)$

The derivative of the exponential is $\frac{d}{du} (e^u) = e^u$:

$$\left(\frac{d}{du} (e^u) \right) \frac{d}{dx} (2x \ln (\sin (e^x))) = (e^u) \frac{d}{dx} (2x \ln (\sin (e^x)))$$

Return to the old variable:

$$e^{(u)} \frac{d}{dx} (2x \ln (\sin (e^x))) = e^{(2x \ln (\sin (e^x)))} \frac{d}{dx} (2x \ln (\sin (e^x)))$$

Apply the constant multiple rule $\frac{d}{dx} (cf(x)) = c \frac{d}{dx} (f(x))$ with $c = 2$ and $f(x) = x \ln (\sin (e^x))$:

$$e^{2x \ln (\sin (e^x))} \left(\frac{d}{dx} (2x \ln (\sin (e^x))) \right) = e^{2x \ln (\sin (e^x))} \left(2 \frac{d}{dx} (x \ln (\sin (e^x))) \right)$$

Apply the product rule $\frac{d}{dx} (f(x)g(x)) = \frac{d}{dx} (f(x)) g(x) + f(x) \frac{d}{dx} (g(x))$ with $f(x) = x$ and $g(x) = \ln (\sin (e^x))$:

$$2e^{2x \ln (\sin (e^x))} \left(\frac{d}{dx} (x \ln (\sin (e^x))) \right) = 2e^{2x \ln (\sin (e^x))} \left(\frac{d}{dx} (x) \ln (\sin (e^x)) + x \frac{d}{dx} (\ln (\sin (e^x))) \right)$$

The function $\ln (\sin (e^x))$ is the composition $f(g(x))$ of two functions $f(u) = \ln (u)$ and $g(x) = \sin (e^x)$.

Apply the chain rule $\frac{d}{dx} (f(g(x))) = \frac{d}{du} (f(u)) \frac{d}{dx} (g(x))$:

$$2 \left(x \left(\frac{d}{dx} (\ln (\sin (e^x))) \right) + \ln (\sin (e^x)) \frac{d}{dx} (x) \right) e^{2x \ln (\sin (e^x))} = 2 \left(x \left(\frac{d}{du} (\ln (u)) \frac{d}{dx} (\sin (e^x)) \right) + \ln (\sin (e^x)) \frac{d}{dx} (x) \right) e^{2x \ln (\sin (e^x))}$$

The derivative of the natural logarithm is $\frac{d}{du} (\ln (u)) = \frac{1}{u}$:

$$2 \left(x \left(\frac{d}{du} (\ln (u)) \right) \frac{d}{dx} (\sin (e^x)) + \ln (\sin (e^x)) \frac{d}{dx} (x) \right) e^{2x \ln (\sin (e^x))} = 2 \left(x \left(\frac{1}{u} \right) \frac{d}{dx} (\sin (e^x)) + \ln (\sin (e^x)) \frac{d}{dx} (x) \right) e^{2x \ln (\sin (e^x))}$$

Return to the old variable:

$$2 \left(\frac{x \frac{d}{dx} (\sin (e^x))}{(u)} + \ln (\sin (e^x)) \frac{d}{dx} (x) \right) e^{2x \ln (\sin (e^x))} = 2 \left(\frac{x \frac{d}{dx} (\sin (e^x))}{(\sin (e^x))} + \ln (\sin (e^x)) \frac{d}{dx} (x) \right) e^{2x \ln (\sin (e^x))}$$

The function $\sin (e^x)$ is the composition $f(g(x))$ of two functions $f(u) = \sin (u)$ and $g(x) = e^x$.

Apply the chain rule $\frac{d}{dx} (f(g(x))) = \frac{d}{du} (f(u)) \frac{d}{dx} (g(x))$:

$$2 \left(\frac{x \left(\frac{d}{dx} (\sin(e^x)) \right)}{\sin(e^x)} + \ln(\sin(e^x)) \frac{d}{dx} (x) \right) e^{2x \ln(\sin(e^x))} =$$

$$2 \left(\frac{x \left(\frac{d}{du} (\sin(u)) \right) \frac{d}{dx} (e^x)}{\sin(e^x)} + \ln(\sin(e^x)) \frac{d}{dx} (x) \right) e^{2x \ln(\sin(e^x))}$$

The derivative of the sine is $\frac{d}{du} (\sin(u)) = \cos(u)$:

$$2 \left(\frac{x \left(\frac{d}{du} (\sin(u)) \right) \frac{d}{dx} (e^x)}{\sin(e^x)} + \ln(\sin(e^x)) \frac{d}{dx} (x) \right) e^{2x \ln(\sin(e^x))} =$$

$$2 \left(\frac{x (\cos(u)) \frac{d}{dx} (e^x)}{\sin(e^x)} + \ln(\sin(e^x)) \frac{d}{dx} (x) \right) e^{2x \ln(\sin(e^x))}$$

Return to the old variable:

$$2 \left(\frac{x \cos((u)) \frac{d}{dx} (e^x)}{\sin(e^x)} + \ln(\sin(e^x)) \frac{d}{dx} (x) \right) e^{2x \ln(\sin(e^x))} =$$

$$2 \left(\frac{x \cos((e^x)) \frac{d}{dx} (e^x)}{\sin(e^x)} + \ln(\sin(e^x)) \frac{d}{dx} (x) \right) e^{2x \ln(\sin(e^x))}$$

The derivative of the exponential is $\frac{d}{dx} (e^x) = e^x$:

$$2 \left(\frac{x \cos(e^x) \left(\frac{d}{dx} (e^x) \right)}{\sin(e^x)} + \ln(\sin(e^x)) \frac{d}{dx} (x) \right) e^{2x \ln(\sin(e^x))} =$$

$$2 \left(\frac{x \cos(e^x) (e^x)}{\sin(e^x)} + \ln(\sin(e^x)) \frac{d}{dx} (x) \right) e^{2x \ln(\sin(e^x))}$$

Apply the power rule $\frac{d}{dx} (x^n) = nx^{n-1}$ with $n = 1$, in other words, $\frac{d}{dx} (x) = 1$:

$$2 \left(\frac{x e^x \cos(e^x)}{\sin(e^x)} + \ln(\sin(e^x)) \left(\frac{d}{dx} (x) \right) \right) e^{2x \ln(\sin(e^x))} =$$

$$2 \left(\frac{x e^x \cos(e^x)}{\sin(e^x)} + \ln(\sin(e^x)) (1) \right) e^{2x \ln(\sin(e^x))}$$

Simplify:

$$2 \left(\frac{x e^x \cos(e^x)}{\sin(e^x)} + \ln(\sin(e^x)) \right) e^{2x \ln(\sin(e^x))} =$$

$$\frac{2(x e^x + \ln(\sin(e^x)) \tan(e^x)) e^{2x \ln(\sin(e^x))}}{\tan(e^x)}$$

Thus, $\frac{d}{dx} (\sin^{2x}(e^x)) = \frac{2(x e^x + \ln(\sin(e^x)) \tan(e^x)) e^{2x \ln(\sin(e^x))}}{\tan(e^x)}$.

4. Find the open intervals on which the function $f(x) = x^4 - 4x^3 + 4x^2 + 3$ is increasing and decreasing. Also find the relative extrema. [8]

Solution

See full solution using the following online solvers:

[Find maximum and minimum value of \$y=x^3+6x^2-15x+7\$ calculator](#)

[Extreme Points Calculator - Find Extreme and Saddle Points](#)

[Derivative calculator](#)

[Derivative Calculator - eMathHelp](#)

[Function Calculator - eMathHelp](#)

[Evaluate Calculator - eMathHelp](#)

To find critical points, first compute $f'(x)$

<https://www.emathhelp.net/calculators/calculus-1/derivative-calculator/?f=x%5E4+-+4x%5E3+%2B4x%5E2+%2B3+&order=1&var=x&p=>

The derivative of a sum/difference is the sum/difference of derivatives:

$$\left(\frac{d}{dx} (x^4 - 4x^3 + 4x^2 + 3) \right) = \left(\frac{d}{dx} (x^4) - \frac{d}{dx} (4x^3) + \frac{d}{dx} (4x^2) + \frac{d}{dx} (3) \right)$$

The derivative of a constant is 0:

$$\left(\frac{d}{dx} (3) \right) + \frac{d}{dx} (4x^2) - \frac{d}{dx} (4x^3) + \frac{d}{dx} (x^4) = (0) + \frac{d}{dx} (4x^2) - \frac{d}{dx} (4x^3) + \frac{d}{dx} (x^4)$$

Apply the power rule $\frac{d}{dx} (x^n) = nx^{n-1}$ with $n = 4$:

$$\left(\frac{d}{dx} (x^4) \right) + \frac{d}{dx} (4x^2) - \frac{d}{dx} (4x^3) = (4x^3) + \frac{d}{dx} (4x^2) - \frac{d}{dx} (4x^3)$$

Apply the constant multiple rule $\frac{d}{dx} (cf(x)) = c\frac{d}{dx} (f(x))$ with $c = 4$ and $f(x) = x^3$:

$$4x^3 - \left(\frac{d}{dx} (4x^3) \right) + \frac{d}{dx} (4x^2) = 4x^3 - \left(4\frac{d}{dx} (x^3) \right) + \frac{d}{dx} (4x^2)$$

Apply the constant multiple rule $\frac{d}{dx}(cf(x)) = c\frac{d}{dx}(f(x))$ with $c = 4$ and $f(x) = x^2$:

$$4x^3 + \left(\frac{d}{dx}(4x^2)\right) - 4\frac{d}{dx}(x^3) = 4x^3 + \left(4\frac{d}{dx}(x^2)\right) - 4\frac{d}{dx}(x^3)$$

Apply the power rule $\frac{d}{dx}(x^n) = nx^{n-1}$ with $n = 2$:

$$4x^3 + 4\left(\frac{d}{dx}(x^2)\right) - 4\frac{d}{dx}(x^3) = 4x^3 + 4(2x) - 4\frac{d}{dx}(x^3)$$

Apply the power rule $\frac{d}{dx}(x^n) = nx^{n-1}$ with $n = 3$:

$$4x^3 + 8x - 4\left(\frac{d}{dx}(x^3)\right) = 4x^3 + 8x - 4(3x^2)$$

Simplify:

$$4x^3 - 12x^2 + 8x = 4x(x - 2)(x - 1)$$

$$\text{Thus, } \frac{d}{dx}(x^4 - 4x^3 + 4x^2 + 3) = 4x(x - 2)(x - 1).$$

Now, we'll find Critical Points by solving the equation:

$$f'(x) = 0$$

$$\Rightarrow 4x(x - 2)(x - 1) = 0$$

$$\text{So, } x = 0 \text{ or } (x - 2) = 0 \Rightarrow x = 2 \text{ or } (x - 1) = 0 \Rightarrow x = 1$$

And, $f'(x) = 4x(x - 2)(x - 1)$ exists everywhere.

Therefore, the critical points of $x^4 - 4x^3 + 4x^2 + 3$ are $x = 0$ or $x = 2$ or $x = 1$

Now:

Open Interval	Test point	Sign of $f'(x) = 4x(x - 2)(x - 1)$	Conclusion
$(-\infty, 0)$ $= \{x \in \mathbb{R} x < 0\}$	-1	-	f is decreasing
$(0, 1)$ $= \{x \in \mathbb{R} 0 < x < 1\}$	0.5	+	f is increasing
$(1, 2)$ $= \{x \in \mathbb{R} 1 < x < 2\}$	1.5	-	f is decreasing
$(2, \infty)$ $= \{x \in \mathbb{R} x > 2\}$	3	+	f is increasing

Now, we'll find $f''(x)$:

<https://www.emathhelp.net/calculators/calculus-1/derivative-calculator/?f=x%5E4+-+4x%5E3+%2B4x%5E2%2B3&order=2&var=&p=>

Next, $\frac{d^2}{dx^2} (x^4 - 4x^3 + 4x^2 + 3) = \frac{d}{dx} (4x(x-2)(x-1))$

Apply the constant multiple rule $\frac{d}{dx} (cf(x)) = c \frac{d}{dx} (f(x))$ with $c = 4$ and $f(x) = x(x-2)(x-1)$:

$$\left(\frac{d}{dx} (4x(x-2)(x-1)) \right) = \left(4 \frac{d}{dx} (x(x-2)(x-1)) \right)$$

Apply the product rule $\frac{d}{dx} (f(x)g(x)) = \frac{d}{dx} (f(x))g(x) + f(x)\frac{d}{dx} (g(x))$ with $f(x) = x$ and $g(x) = (x-2)(x-1)$:

$$4 \left(\frac{d}{dx} (x(x-2)(x-1)) \right) = 4 \left(\frac{d}{dx} (x)(x-2)(x-1) + x \frac{d}{dx} ((x-2)(x-1)) \right)$$

Apply the product rule $\frac{d}{dx} (f(x)g(x)) = \frac{d}{dx} (f(x))g(x) + f(x)\frac{d}{dx} (g(x))$ with $f(x) = x-1$ and $g(x) = x-2$:

$$4x \left(\frac{d}{dx} ((x-2)(x-1)) \right) + 4(x-2)(x-1) \frac{d}{dx} (x) =$$

$$4x \left(\frac{d}{dx} (x-1)(x-2) + (x-1) \frac{d}{dx} (x-2) \right) + 4(x-2)(x-1) \frac{d}{dx} (x)$$

The derivative of a sum/difference is the sum/difference of derivatives:

$$4x \left((x-2) \left(\frac{d}{dx} (x-1) \right) + (x-1) \frac{d}{dx} (x-2) \right) + 4(x-2)(x-1) \frac{d}{dx} (x) =$$

$$4x \left((x-2) \left(\frac{d}{dx} (x) - \frac{d}{dx} (1) \right) + (x-1) \frac{d}{dx} (x-2) \right) + 4(x-2)(x-1) \frac{d}{dx} (x)$$

The derivative of a constant is 0:

$$4x \left((x-2) \left(- \left(\frac{d}{dx} (1) \right) + \frac{d}{dx} (x) \right) + (x-1) \frac{d}{dx} (x-2) \right) +$$

$$4(x-2)(x-1) \frac{d}{dx} (x) = 4x \left((x-2) \left(-(0) + \frac{d}{dx} (x) \right) + (x-1) \frac{d}{dx} (x-2) \right) +$$

$$4(x-2)(x-1) \frac{d}{dx} (x)$$

Apply the power rule $\frac{d}{dx}(x^n) = nx^{n-1}$ with $n = 1$, in other words, $\frac{d}{dx}(x) = 1$:

$$4x \left((x-2) \left(\frac{d}{dx}(x) \right) + (x-1) \frac{d}{dx}(x-2) \right) + 4(x-2)(x-1) \left(\frac{d}{dx}(x) \right) =$$

$$4x \left((x-2)(1) + (x-1) \frac{d}{dx}(x-2) \right) + 4(x-2)(x-1)(1)$$

The derivative of a sum/difference is the sum/difference of derivatives:

$$4x \left(x + (x-1) \left(\frac{d}{dx}(x-2) \right) - 2 \right) + 4(x-2)(x-1) =$$

$$4x \left(x + (x-1) \left(\frac{d}{dx}(x) - \frac{d}{dx}(2) \right) - 2 \right) + 4(x-2)(x-1)$$

Apply the power rule $\frac{d}{dx}(x^n) = nx^{n-1}$ with $n = 1$, in other words, $\frac{d}{dx}(x) = 1$:

$$4x \left(x + (x-1) \left(\left(\frac{d}{dx}(x) \right) - \frac{d}{dx}(2) \right) - 2 \right) + 4(x-2)(x-1) =$$

$$4x \left(x + (x-1) \left((1) - \frac{d}{dx}(2) \right) - 2 \right) + 4(x-2)(x-1)$$

The derivative of a constant is 0:

$$4x \left(x + \left(1 - \left(\frac{d}{dx}(2) \right) \right) (x-1) - 2 \right) + 4(x-2)(x-1) =$$

$$4x \left(x + (1 - (0)) (x-1) - 2 \right) + 4(x-2)(x-1)$$

Simplify:

$$4x(2x-3) + 4(x-2)(x-1) = 12x^2 - 24x + 8$$

$$\text{Thus, } \frac{d}{dx}(4x(x-2)(x-1)) = 12x^2 - 24x + 8.$$

$$\text{Therefore, } \frac{d^2}{dx^2}(x^4 - 4x^3 + 4x^2 + 3) = 12x^2 - 24x + 8.$$

ANSWER

$$\frac{d^2}{dx^2}(x^4 - 4x^3 + 4x^2 + 3) = 12x^2 - 24x + 8 \quad \mathbf{A}$$

Evaluating $f''(x)$ at critical point $x = 0$: $f''(0) = 12(0)^2 - 24 \cdot (0) + 8 = 0 - 0 + 8 = 8$

Since, $f''(0) > 0$. So, $f(x)$ has a minima at $x = 0$.

Evaluating $f''(x)$ at critical point $x = 1$:

$$f''(1) = 12(1)^2 - 24 \cdot (1) + 8 = 12 - 24 + 8 = -4$$

Since, $f''(1) < 0$. So, $f(x)$ has a maxima at $x = 1$.

Evaluating $f''(x)$ at critical point $x = 2$:

$$f''(2) = 12(2)^2 - 24 \cdot (2) + 8 = 48 - 48 + 8 = 8$$

Since, $f''(2) > 0$. So, $f(x)$ has a minima at $x = 2$.

5. Verify the statement of the Mean Value Theorem for the function [8]

$$f(x) = x^3 - x^2 + 9x - 2$$

over the interval $[1,4]$, and find all values of 'c' in that interval that satisfy the conclusion of the theorem.

Solution

See full solution here:

<https://www.emathhelp.net/calculators/calculus-1/mean-value-theorem-calculator/?f=x%5E3+-x%5E2%2B9x+-2&a=1&b=4>

The Mean Value Theorem states that for a continuous and differentiable function $f(x)$ on the interval $[a, b]$ there exists such number c from the interval (a, b) , that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

First, evaluate the function at the endpoints of the interval:

$$f(4) = 82$$

$$f(1) = 7$$

Next, find the derivative: $f'(c) = 3c^2 - 2c + 9$ (for steps, see [derivative calculator](#)).

$$\text{Form the equation: } 3c^2 - 2c + 9 = \frac{(82) - (7)}{(4) - (1)}$$

$$\text{Simplify: } 3c^2 - 2c + 9 = 25$$

$$\text{Solve the equation on the given interval: } c = \frac{8}{3}$$

$$\text{Answer: } \frac{8}{3} \approx 2.66666666666667$$

6. Find the Maclaurin polynomials P_0, P_1, P_2, P_3 of the function $f(x) = \ln\left(\frac{1+x}{1-x}\right)$. [8]

Hint: Use the formula $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$.

Solution

See full solution here:

<https://www.emathhelp.net/calculators/calculus-1/taylor-and-maclaurin-series-calculator/?f=ln+%28%281%2Bx%29+%2F%281-x%29%29&p=0&n=3&v=>

Using the fact that $P_n = P_{n-1} + n - \text{th order terms}$, We can get P_0, P_1, P_2, P_3 sequentially.

Your input: calculate the Taylor (Maclaurin) series of $\ln\left(\frac{x+1}{1-x}\right)$ up to $n = 3$

A Maclaurin series is given by $f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} x^k$

In our case, $f(x) \approx P(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} x^k = \sum_{k=0}^3 \frac{f^{(k)}(a)}{k!} x^k$

So, what we need to do to get the desired polynomial is to calculate the derivatives, evaluate them at the given point, and plug the results into the given formula.

$$f^{(0)}(x) = f(x) = \ln\left(\frac{x+1}{1-x}\right)$$

Evaluate the function at the point: $f(0) = 0$

- Find the 1st derivative: $f^{(1)}(x) = (f^{(0)}(x))' = \left(\ln\left(\frac{x+1}{1-x}\right)\right)' = -\frac{2}{x^2-1}$ (steps can be seen [here](#)).

Evaluate the 1st derivative at the given point: $(f(0))' = 2$

- Find the 2nd derivative: $f^{(2)}(x) = (f^{(1)}(x))' = \left(-\frac{2}{x^2-1}\right)' = \frac{4x}{(x^2-1)^2}$ (steps can be seen [here](#)).

Evaluate the 2nd derivative at the given point: $(f(0))'' = 0$

- Find the 3rd derivative: $f^{(3)}(x) = (f^{(2)}(x))' = \left(\frac{4x}{(x^2-1)^2}\right)' = \frac{4(-3x^2-1)}{(x^2-1)^3}$ (steps can be seen [here](#)).

Evaluate the 3rd derivative at the given point: $(f(0))''' = 4$

Now, use the calculated values to get a polynomial:

$$f(x) \approx \frac{0}{0!}x^0 + \frac{2}{1!}x^1 + \frac{0}{2!}x^2 + \frac{4}{3!}x^3$$

Finally, after simplifying we get the final answer:

$$f(x) \approx P(x) = 2x + \frac{2}{3}x^3$$

Answer: the Taylor (Maclaurin) series of $\ln\left(\frac{x+1}{1-x}\right)$ up to $n = 3$ is

$$\ln\left(\frac{x+1}{1-x}\right) \approx P(x) = 2x + \frac{2}{3}x^3$$